

Comparing Groups with Two-Way Tables

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

- | | |
|---|--|
| 1. 20 | 7. walk |
| 2. 28 | 8. the Year 8 fraction = $\frac{3}{10}$, the Year 7 fraction = $\frac{1}{10}$, the comparison symbol = $>$ |
| 3. $\frac{7}{16}$ | 9. 57.5 |
| 4. the under-30 fraction = $\frac{9}{16}$, the 30-and-over fraction = $\frac{7}{12}$, the comparison symbol = $<$ | 10. the Year 7 percentage = 75, the Year 9 percentage = 40, — |
| 5. 84 | |
| 6. the percentage for non-attenders = 60, — | |
-

What is verified

15 of 17 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 6, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.SP.A.4 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 17 tagged checks.

1. Sport and walking to school: how many play basketball and do not walk?

Solution: Find the Basketball row, then run across to the “Does not walk” column. The cell where they meet counts the students in both categories at once. 20

2. Same survey: how many students walk to school altogether?

Solution: The “Walks” column holds 18 footballers and 10 basketball players. Adding down the column gives the column total: $18 + 10$. It matches the total already printed in the bottom row, which is a useful check. 28

3. Cats and dogs: what fraction of the under-30 group prefers cats?

Solution: The question is about the under-30 group only, so the denominator is that row’s total, 80, not the grand total of 140. The numerator is the cell where the row meets the cats column: $\frac{35}{80}$. Dividing both parts by 5 simplifies it. $\frac{7}{16}$

4. Cats and dogs: compare the two groups' dog-preferring fractions.

Solution: Each fraction uses its own row total.

$$\text{under 30 : } \frac{45}{80} = \frac{9}{16} = 0.5625, \quad 30 \text{ and over : } \frac{35}{60} = \frac{7}{12} = 0.5833\dots$$

The second is larger, even though the first came from more people — which is exactly why the raw counts 45 and 35 cannot be compared directly.

$$\boxed{\frac{9}{16} \quad \text{and} \quad \frac{7}{12}, \quad \text{so} \quad \frac{9}{16} < \frac{7}{12}}$$

5. Homework club: what percent of the club attenders passed?

Solution: The attenders row totals 50 and 42 of them passed, so the row percentage is $\frac{42}{50} \times 100 = 0.84 \times 100$. $\boxed{84}$

6. Homework club: the non-attenders' pass percentage, and whether there is an association.

(a) **Solution:** $\frac{30}{50} \times 100$. $\boxed{60}$

(b) **Description (instructor-judged).** A correct response compares the two row percentages: 84 percent of the club attenders passed against 60 percent of the non-attenders, a gap of 24 percentage points, so attending the club and passing do appear to be associated in this school. Comparing the raw counts 42 and 30 without noticing that each row holds 50 students earns partial credit only — here the rows happen to be equal, but that is not something to assume.

7. Travel survey: which way of travelling is used by the most students overall?

Solution: The bottom row holds the column totals: 42 by bus, 54 walking, 24 cycling. The largest is walking. $\boxed{\text{walk}}$

8. Travel survey: compare the cycling fractions of the two year groups.

Solution: Each year group has a row total of 60.

$$\text{Year 8 : } \frac{18}{60} = \frac{3}{10}, \quad \text{Year 7 : } \frac{6}{60} = \frac{1}{10}$$

Year 8 cycles three times as often as Year 7.

$$\boxed{\frac{3}{10} \quad \text{and} \quad \frac{1}{10}, \quad \text{so} \quad \frac{3}{10} > \frac{1}{10}}$$

9. School lunch: what percent of all students surveyed liked it?

Solution: “All students” means the grand total, 120, and 69 of them liked it: $\frac{69}{120} \times 100 = 0.575 \times 100$. 57.5

10. School lunch: the two year groups’ percentages, and whether opinion is associated with year group.

Solution: Each year group has 60 students, so divide by 60 and multiply by 100: $\frac{45}{60} \times 100$ and $\frac{24}{60} \times 100$.

Year 7 = 75	Year 9 = 40
-------------	-------------

(c) Judgement (instructor-judged). A correct response uses the two percentages: 75 percent of Year 7 liked the lunch against 40 percent of Year 9, a gap far too large to dismiss, so year group and opinion do appear associated in this survey. A full answer also explains what the percentages add: they express each group’s opinion as a share of its own row, which is what makes two groups comparable even when the rows hold different numbers of students. A response built on the raw counts alone earns partial credit.